

Reflective Journal – Lab 10: Video Analysis and Object Tracking

Overview

Lab 10 was where things finally clicked for me about the difference between detection and tracking. Before this, I thought bounding boxes were enough, but once I saw how identity continuity matters across frames, it made sense why tracking is its own challenge. Using YOLO with ByteTrack and BoT-SORT gave me a chance to see how trackers try to connect detections over time — and also how often they mess up.

Key Learnings

- **Detection vs. Tracking:** Detection just labels each frame, but tracking adds IDs so you can follow the same person across frames. That’s the piece that makes counting unique individuals possible.
- **Failure Modes:** I ran into ID switches and phantom tracks constantly. It was frustrating to watch the tracker “invent” people or swap IDs mid-video. It made me realize how fragile these systems can be.
- **Motion vs. Appearance Cues:** ByteTrack was fast but broke down when people overlapped or stopped moving. BoT-SORT was better at keeping identities straight because it used appearance features, but it was slower. That trade-off between speed and robustness is real.
- **Bandwidth and Compression:** Doing the math on uncompressed video bandwidth was eye-opening. The raw data requirements are ridiculous, and compression is the only reason large-scale monitoring is even possible.

Application to Practice

I can see how this applies in industries like transportation, where tracking vehicles across intersections matters, or retail, where stores want to count unique customers. Even in sports analysis, tracking players across frames is critical. The failures I saw — ID switches, phantom tracks — would directly affect accuracy in those contexts. It made me think about how much engineering effort goes into making these systems reliable enough for real use.

Reflection

Honestly, this lab was frustrating at times. Watching the tracker lose identities or invent phantom ones made me question how “smart” these systems really are. But that frustration was useful — it forced me to pay closer attention and diagnose why things were failing. I realized that tracking isn’t just a technical add-on to detection; it’s a whole separate problem with its own trade-offs.

I also found the bandwidth exercise surprisingly interesting. I hadn’t thought about how much raw video data adds up, and it made me appreciate why compression and efficiency matter so much in real deployments.

Overall, Lab 10 helped me connect theory to practice. It showed me that trackers are essential but imperfect, and that engineers have to balance speed, accuracy, and resource limits. More than anything, it reminded me that building computer vision systems isn't just about getting bounding boxes — it's about designing for reliability in messy, real-world conditions.